## Performance Work Statement

## Addition of TRACER & Similar Modem Support by the TFRMS as External References for use with the TFRMS Clock Ensemble

**BACKGROUND:**

The U.S. Naval Observatory (USNO) is responsible for the Critical Time Dissemination (CTD) project driven by the Department of War (DoW) Chief Information Office (CIO) and created by the USNO partnered with the Defense Information Systems Agency (DISA). The project utilizes the Time and Frequency Reference Measurement Systems (TFRMS),a rack mounted unit with various external interface cards, oscillators, and GPS Receivers. Software upgrades are needed for the interface cards to effectively use external time transfer signals for improved performance. The performance of the CTD system and its future is tracked by the DoW CIO and across the DoW and IC. It is critical that the software is upgraded and deployed to ensure the best performance around the world.

**SCOPE:**

The below items represent the minimum requirements of the Government.

**REQUIREMENTS:**

1. Modify the TFRMS software (only) to accept input from the new TRACER Modem as an external reference. The solution provided to support the TRACER modem will also support other types of external reference sources in addition to the TRACER modem that provide the same input data format and data content to the TFRMS (hereinafter the TRACER and other similar devices will be referred to as “Modem” or “Modem(s)”). Data commands, data format, and data content will be approved by the Government Point(s) of Contact listed below.

2. The TFRMS will provide 1PPS, 10MHz (Sinewave) and IRIG (B127) as reference inputs to the Modems from the existing TFRMS hardware interfaces. No new hardware will be added to the TFRMS to accomplish the task of supporting the new external references. The TFRMS will not provide L-Band Rx input to the Modem(s), nor will the TFRMS accept or use the “Future Feature” outputs or the L-Band Tx output from the Modem(s).

3. The Modems will output offset data with picosecond resolution once per second via a 100Mbps Ethernet interface for 8-channels of source data (**Threshold**). Data from a second Modem, for an additional 8-channels, may be accommodated by the updated TFRMS software, for a total of 16 channels (**Objective**). All channels will be prioritized and selected for inclusion, included, selected for removal, removed, or excluded as determined by the control software that runs on an external PC. The PC will push the offset data via Ethernet to the TFRMS and the TFRMS will connect to the PC via TCP/IP.

4. The external PC and attached Modems will be connected to the 10/100Mbps port of the TFRMS, which supports NTP Server Mode, via an external Ethernet Hub. Data for each channel will be input to the TFRMS at a rate of once per second, at a transmit rate of 1kbps.

5. A single-state Kalman filter will be applied to each of the 8 (**Threshold**) or 16 (**Objective**) data input streams from the Modems using the processor in the GTF Module. The single-stated Kalman filter will be the same as currently employed in the TFRMS for use with the Ensemble.

6. The TFRMS Ensemble software will be modified to accommodate 8 (**Threshold**) or 16 (**Objective**) data input streams from the Modems, to include priority and weighting.

7. The TFRMS GTF Intercommunication software will be modified to allow for seamless transition between the GTFs should a GTF Module fail or be removed during operation.

## REPORTS TO USNO:

**Acceptance and Quality Assurance**
The contractor shall present final report(s) to the USNO technical lead for acceptance. The USNO technical lead shall address any issues that may arise during the execution of this contract.

**Dissemination control of output products such as documents:** Technical report(s) shall be considered "DISTRIBUTION D: DoW AND DoW CONTRACTORS ONLY," and shall not be further disseminated without written permission of USNO. Contractor shall comply with all relevant USNO, Navy, DoW dissemination, security and information assurance rules.

## PERIOD OF PERFORMANCE:

The period of performance is 30 weeks from the date of Award.